The Selection Rule

For rigid diatomic rotators:

$$\Delta J = \pm 1$$

The rotational spectrum is observed only if the molecule is asymmetric (heteronuclear), eg. HCl, CO.

The rotational spectrum is not observed if the molecule is homonuclear, eg. N_2, O_2.

The intensities of spectral lines

$$\bar{\nu}_{J \to J+1} = 2B(J+1)$$

Requirement:
A knowledge of relative probabilities of transition between the various energy levels.

Whether
a molecule has more or less chance of making the transition J=0 $\rightarrow$ J=1 than the transition J=1 $\rightarrow$ J=2.

Remember
$\Delta J = \pm 2, \pm 3$, etc. are forbidden (**transition probability is zero**).

To have allowed transitions, the **transition moment integral should be non-zero**. This is possible if $\Delta J = \pm 1$.

$$\boxed{\vec{M}_{21} = \int \Psi_2 \vec{\mu} \Psi_1 d\tau}$$

where, The Ψ_1 and Ψ_2 are two different eigenstates in one molecule, $\vec{M}_{21}$ is the electric dipole moment operator.

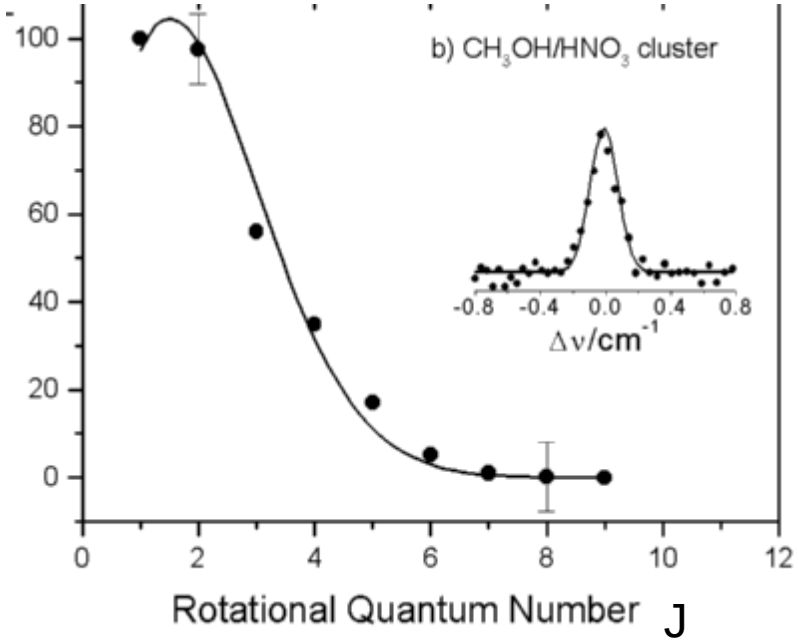
Rotational energy in the lowest level is zero, since J = 0.

If there are N_0 molecules present in this state then the content of any higher state is given by

$$\frac{N_J}{N_0} = \exp\left(-E_J/kT\right) = exp[-BhcJ(J+1)/kT]$$

N_J varies with J.

$exp[-BhcJ(J+1)/kT]$



Rotational state distributions of nascent OH fragments from 193 nm photolysis of HNO_3 in methanol clusters.

Possibility of degeneracy in the energy states

Degeneracy $\rightarrow$ existence of two or more energy states having exactly the same energy.

Diatomic rotator $\rightarrow$ angular momentum dictates the degeneracy.

Rotator → Energy $E = \frac{1}{2} I\omega^2$

Angular momentum $P = I\omega$

I is the moment of inertia, ω is the rotational frequency.

$$P = \sqrt{2EI}$$

We know: $E = \frac{h^2}{8\pi^2 I} J(J+1) \rightarrow 2EI = \frac{h^2}{4\pi^2} J(J+1)$

So: $P = \sqrt{J(J+1)}\frac{h}{2\pi} = \sqrt{J(J+1)}$ units

h/2π → fundamental unit of angular momentum

$$P = \sqrt{J(J+1)}\frac{h}{2\pi} = \sqrt{J(J+1)}\ \text{units}$$

P is quantized.

P is a vector – it has direction as well as magnitude.

The direction of the angular momentum is conventionally taken to be along the x-axis about which rotation occurs.

The number of different directions which an angular momentum vector may take is limited by:

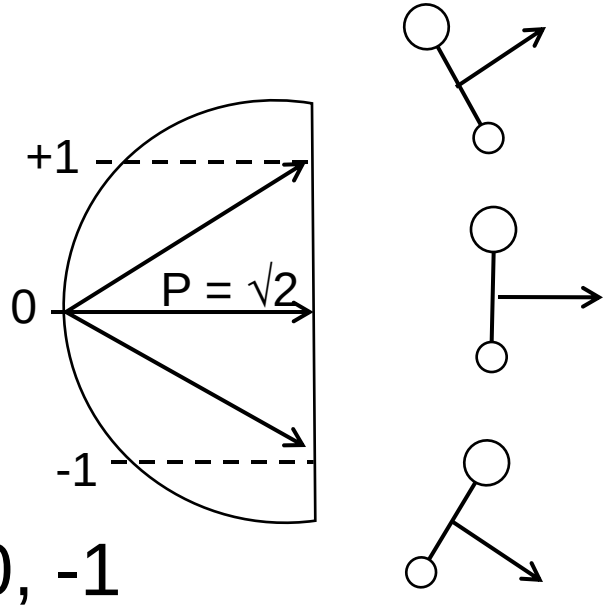
For integral values of the rotational quantum number (J), the angular momentum vector (**P**) may only take up directions such that its components along a given reference direction is zero or an integral multiple of angular momentum units.

$$P = \sqrt{J(J+1)}\frac{h}{2\pi} = \sqrt{J(J+1)}\ \text{units}$$

When J = 1

P = √(1x2) units = √2

P can have three directions → +1, 0, -1

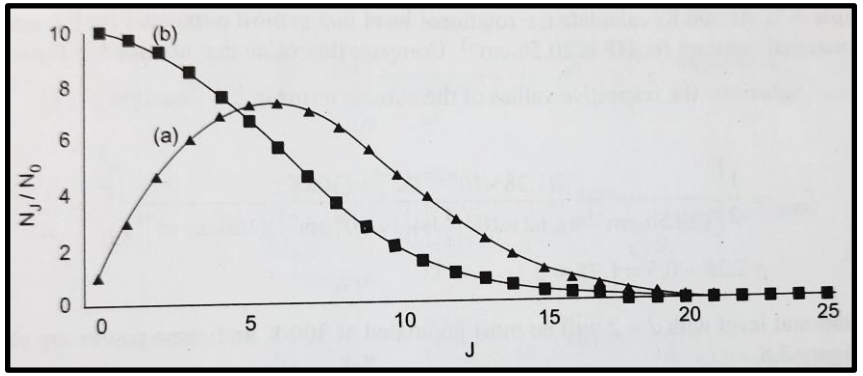


Thus, the angular momentum vector can be oriented in only three directions when J = 1.

All the three rotational directions have the same rotational energy → J = 1 is three-fold degenerate.

Each energy level is (2J+1)-fold degenerate

The total relative population at an energy E_J is



Population $\propto$ $(2J+1)\exp(-E_J/kT)$

Population rises to a maximum and then diminishes with increasing J.

Differentiation of the above equation shows that:

Maximum population: $$J = \sqrt{\frac{kT}{2hcB}} - \frac{1}{2}$$

- Transitions between levels with very low and very high J values will have small intensities
- while the intensity will be maximum at or near the J value given by this equation.

The effect of isotopic substitution

Isotope: An element identical in every way except for its atomic mass.

Replacement of a particular atom in a molecule by its isotope makes the resulting substance chemically identical with the original.

Change will be in the total mass and hence the moment of inertia and B value for the molecule.

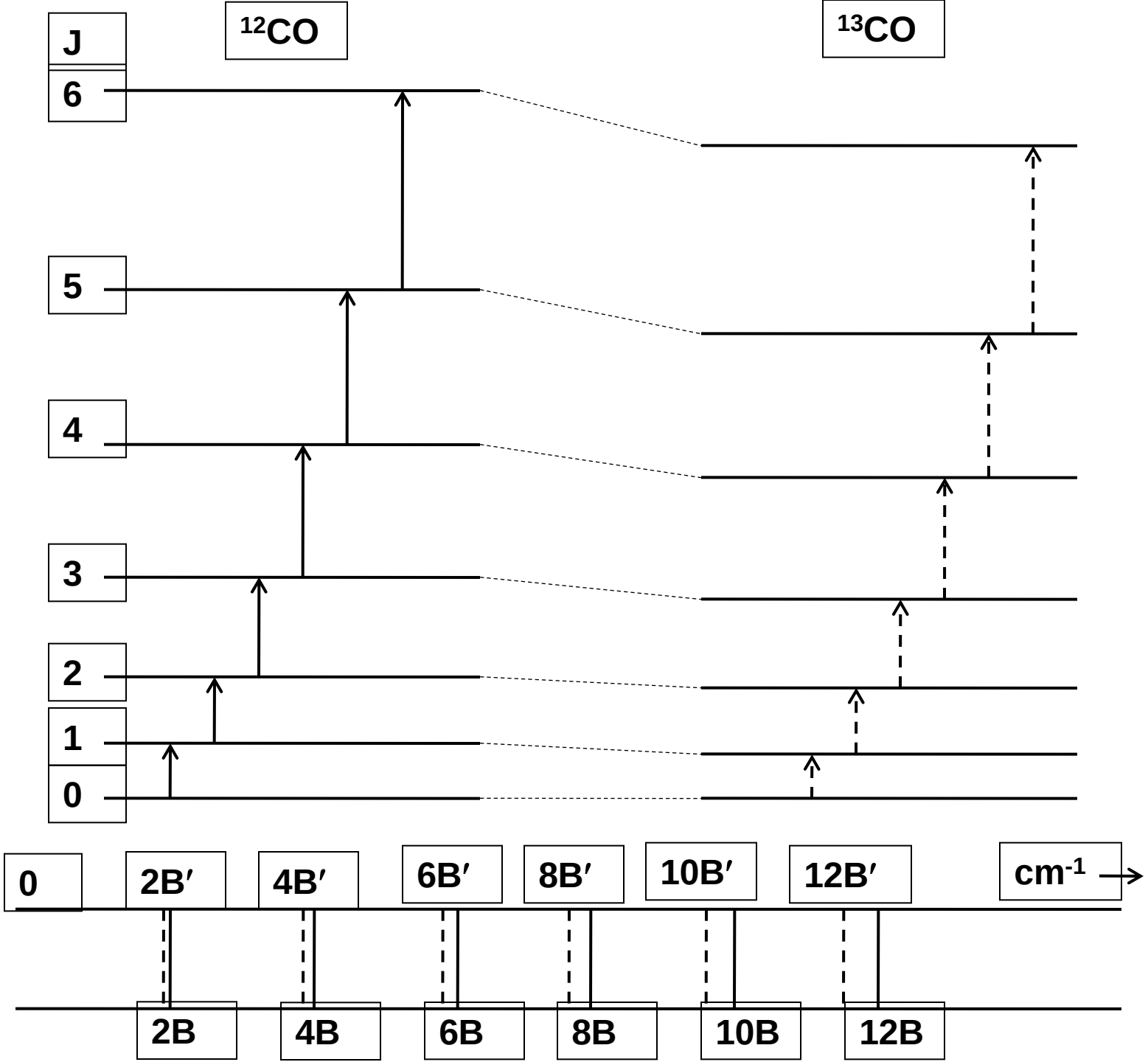
Example: carbon monoxide

$^{12}C^{16}O \rightarrow {}^{13}C^{16}O$

Increase in mass and decrease in the B value. $B = \frac{h}{8\pi^2 Ic}$

i.e., B (for $^{12}C^{16}O$) > B′ (for $^{13}C^{16}O$)

This will affect the rotational energy levels of the molecule.


J
12CO
13CO
6
5
4
3
2
1
0
0
2B′
4B′
6B′
8B′
10B′
12B′
cm-1
2B
4B
6B
8B
10B
12B

Determination of precise atomic weights

Rotational absorption of $^{12}C^{16}O$ is at 3.84235 cm^{-1}.

Rotational absorption of $^{13}C^{16}O$ is at 3.67337 cm^{-1}.

Calculated values of B = 1.92118 cm^{-1}

and B′ = 1.83669 cm^{-1}

Hence, B/B′ = $I'/I = \mu'/\mu = 1.046$

The internuclear distance is considered unchanged by isotopic substitution.

Mass of O = 15.9994 and 12-C = 12.00

$$\frac{\mu'}{\mu} = 1.046 = \frac{15.9994m'}{15.9994 + m'} \times \frac{12 + 15.9994}{12 \times 15.9994}$$

Atomic wt. of13-C (m′) = 13.0007

The Non-rigid Rotator

Bond length increases with J.

Assumption about rigid body is only an approximation.

All bonds are elastic to some extent.

More quickly the diatomic molecule rotates the greater is the centrifugal force tending to move the atoms apart.

Consequences of elasticity

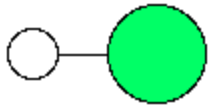
When the bond is elastic, a molecule may have vibrational energy.

The bond will stretch and compress periodically with a certain <u>fundamental frequency</u>.

Depends on the masses of the atoms and the elasticity (force constant) of the bond.

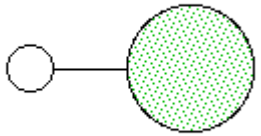
^{1}HF

$\nu = 4139\ cm^{-1}$



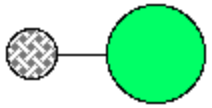
^{1}HCl

$\nu = 2991\ cm^{-1}$



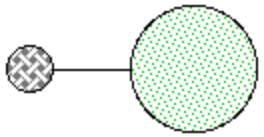
^{2}HF

$\nu = 2998\ cm^{-1}$



^{2}HCl

$\nu = 2145\ cm^{-1}$



If the motion is simple harmonic the force constant is given by

$$k = 4\pi^2 \overline{\omega}^2 c^2 \mu$$

Vibration frequency (cm^{-1})

The variation of B and J is determined by the force constant.

The weaker the bond, the more rapidly will it distort under centrifugal forces.

B and r vary during a vibration.

The measured value of B is an average.

$$B = \frac{h}{8\pi^2 Ic} = \frac{h}{8\pi^2 \mu r^2 c}$$

$$\boxed{B \propto \frac{1}{r^2}}$$

All other quantities are independent of vibration.

Three different sets of values for B and r can be obtained:

<u>At the equilibrium separation</u>:

B_e (rotational const.), r_e (internuclear distance)

<u>In the vibrational ground state</u>:

B_0 (rotational const.), r_0 (average internuclear distance)

<u>At excess vibrational energy</u>:

B_v (rotational const.), r_v (internuclear distance)

Spectrum of a non-rigid rotator

The Schrödinger wave equation for a non-rigid rotator provides the rotational energy levels:

$$E_J = \frac{h^2}{8\pi^2 I} J(J+1) - \frac{h^4}{32\pi^4 I^2 r^2 k} J^2 (J+1)^2 \text{ joules}$$

$$\Downarrow$$

$$\varepsilon_J = \frac{E_J}{hc} = BJ(J+1) - DJ^2(J+1)^2 \text{ cm}^{-1}$$

Rotational constant

Centrifugal Distortion constant

$$\varepsilon_J = \frac{E_J}{hc} = BJ\left(J+1\right) - DJ^2(J+1)^2 \text{ cm}^{-1}$$

For simple harmonic force field

$$\varepsilon_J = \frac{E_J}{hc} = BJ\left(J+1\right) - DJ^2(J+1)^2 + \mathrm{HJ}^3\left(\mathrm{J}+1\right)^3 + \mathrm{KJ}^4\left(\mathrm{J}+1\right)^4 .... \text{ cm}^{-1}$$

For anharmonic force field

H, K, etc. are constants dependent upon the geometry of the molecule.

$$B = \frac{h^2}{8\pi^2 I} \text{ joules} = \frac{h}{8\pi^2 Ic} \text{ cm}^{-1}$$

$$D = \frac{h^4}{32\pi^4 I^2 r^2 k} \text{ joules} = \frac{h^3}{32\pi^4 I^2 r^2 kc} \text{ cm}^{-1}$$

$$\Downarrow$$

$$D = \frac{16B^3\pi^2\mu c^2}{k} = \frac{4B^3}{\overline{\omega}^2}$$

D is very small compared to B.

Selection Rule for non-rigid rotator is
$\Delta J = \pm 1$

$$\varepsilon_{J+1} - \varepsilon_J = \bar{\nu}_J$$

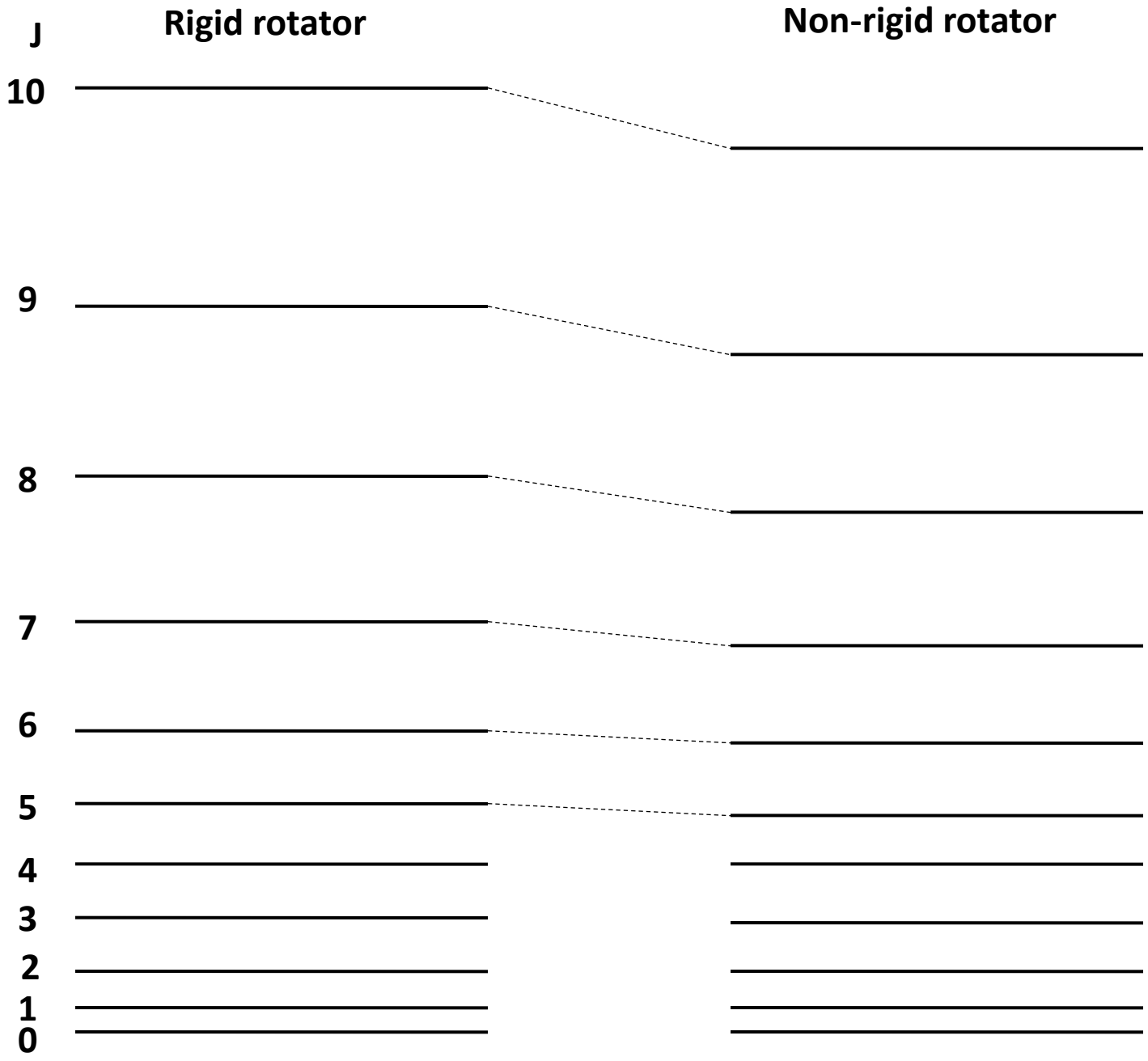
$$= B\left[(J+1)(J+2) - J(J+1)\right] - \mathrm{D}\left[(\mathrm{J}+1)^2(J+2)^2 - J^2(J+1)^2\right]\mathrm{cm}^{-1}$$

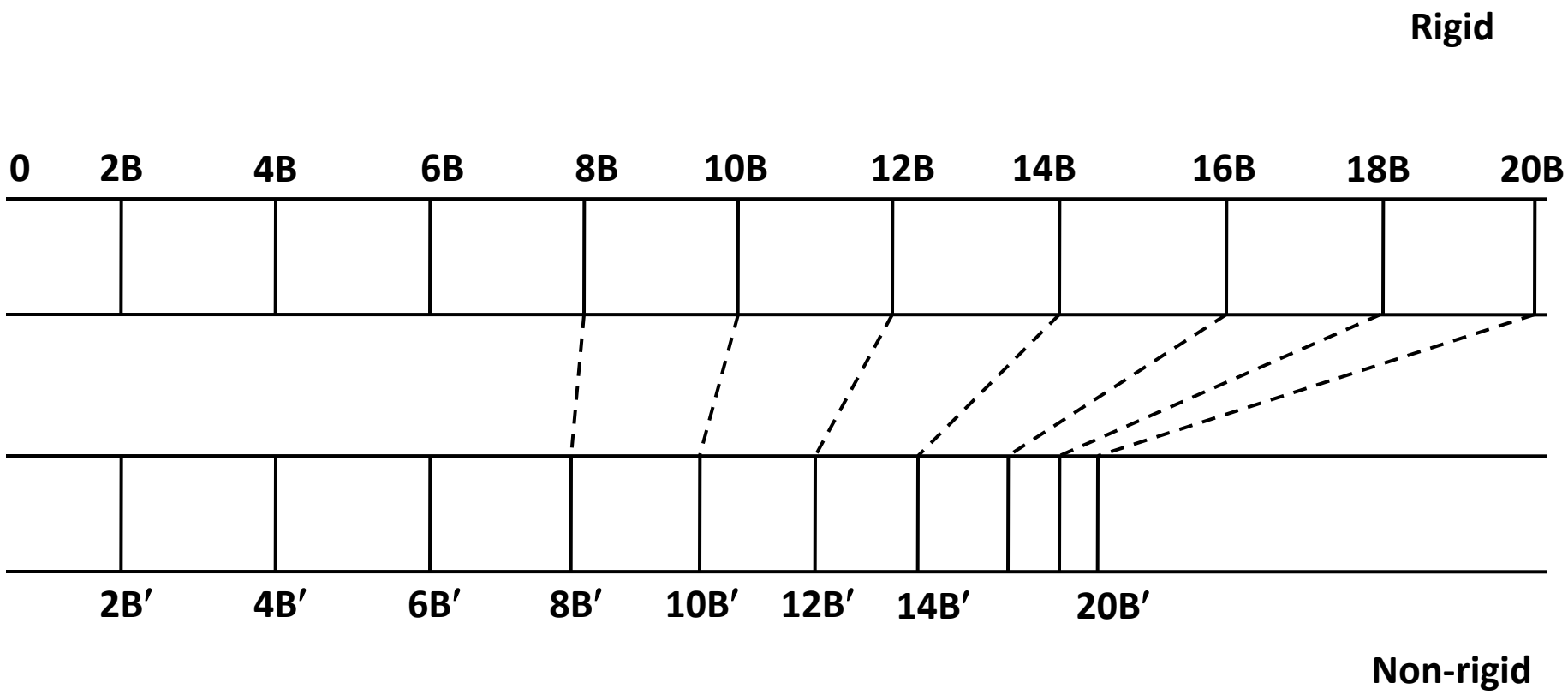
$$= 2B(J+1) - 4D(J+1)^3\ \mathrm{cm}^{-1}$$

The spectrum of the elastic rotor is similar to that of the rigid rotor.

<u>Except</u>:

Each line is displaced slightly to low frequency due to the second term.


J
Rigid rotator
Non-rigid rotator
10
9
8
7
6
5
4
3
2
1
0


Rigid
0
2B
4B
6B
8B
10B
12B
14B
16B
18B
20B
2B′
4B′
6B′
8B′
10B′
12B′
14B′
20B′
Non-rigid

Polyatomic Molecules

Linear Molecules

Examples: Carbon oxysulphide (OCS)
Chloroacetylene (HC≡CCl)

$$I_B = I_C \text{ , } I_A = 0$$

Energy levels

$$\varepsilon_J = BJ(J+1) - DJ^2(J+1)^2 + \ldots\ldots \text{ cm}^{-1}$$

The moment of inertia for the end-over-end rotation for a polyatomic molecule is considerably greater than that of a diatomic molecule.

↓

The B value will be much smaller.

The spectral lines will be more closely spaced.

The molecule must possess a dipole moment to exhibit a spectrum.

OCS will be rotationally active (microwave active) while OCO will not.

$^{16}OC^{18}O$ is microwave inactive.

Isotopic substitution does not lead to a dipole moment.

The bond lengths and atomic charges are unaltered by the substitution.

Symmetric Top Molecule

$$I_B = I_C \neq I_A \qquad I_A \neq 0$$

Consider CH_3F

Two directions of rotation in which the molecule absorbs or emits radiation.

- about the main symmetry axis (C-F bond axis).
- perpendicular to this axis.

Two quantum numbers are necessary:
One for I_A and one for I_B or I_C.

Consider J for the end-over-end rotation of the molecule and K for the rotation about the C-F axis.

On choosing a value for J, K becomes more limited.

Let J = 3.

K cannot be greater than J, since J is the total angular momentum.

So, K = 3, 2, 1 or 0.

K can be negative.

Positive K for clockwise rotation and negative K for anticlockwise rotation of molecules.

So, for a particular J;
K = J, J-1, J-2, …., 0, …., -(J-1), -J

That is 2J+1 values for K.

Rigid Symmetric Top Molecule

Bonds do not stretch under centrifugal forces.

The energy levels are given by:

$$\varepsilon_{J,K} = {E_{J,K}}/{hc} = BJ(J+1) + (A-B)K^2 \text{ cm}^{-1}$$

$$\text{where } B = \frac{h}{8\pi^2 I_B c} \text{ and } A = \frac{h}{8\pi^2 I_A c}$$

The energy depends on $K^2 \rightarrow$ it is immaterial whether the top spins clockwise or anticlockwise.

Selection Rule for Rigid Symmetric Top Molecule

$\Delta J = \pm 1$ and $\Delta K = 0$

The spectrum is given by:

$$\varepsilon_{J+1,K} - \varepsilon_{J,K} = \bar{\nu}_{J,K} = BJ(J+1)(J+2) + (A-B)K^2 - \left[BJ(J+1) + (A-B)K^2\right]$$

$$= 2B(J+1)\,\mathrm{cm}^{-1}$$ Same as linear molecule.

The spectrum is independent of K.

The rotation about the symmetry axis does not change the dipole moment perpendicular to the axis and hence cannot interact with radiation.